ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE D · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Electronics and Digital Concepts for Robotics paper 1 · practice d.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable diagrams.

SECTION A

QUESTION 1

1 1.1 A ✓ 1.2 C ✓ 1.3 B ✓ 1.4 D ✓ 1.5 B ✓ 1.6 A ✓ 1.7 D ✓ 1.8 C ✓ 1.9 A ✓ 1.10 B ✓ (10)

[10]

QUESTION 2

2 2.1 True ✓ 2.2 False ✓ 2.3 False ✓ 2.4 True ✓ 2.5 True ✓ 2.6 True ✓ 2.7 False ✓ 2.8 True ✓ 2.9 False ✓ 2.10 True ✓ (10)

[10]

QUESTION 3

3 3.1 C ✓ 3.2 F ✓ 3.3 G ✓ 3.4 A ✓ 3.5 I ✓ 3.6 E ✓ 3.7 B ✓ 3.8 D ✓ 3.9 H ✓ 3.10 J ✓ (10)

[10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1.1 An emf is produced in a conductor when it cuts magnetic lines of force, or when the flux linking a coil changes ✓ (1)

4.1.2 The direction of an induced current is always such that it opposes the change that causes it ✓ (1)

4.1.3 The rotating part of a DC machine that carries the windings in which the emf is induced ✓ (1)

4.1.4 A segmented copper ring on the shaft that changes the alternating emf in the armature into DC at the brushes (reverses the connections every half turn) ✓ (1)

4.2.1 $I_f = V \div R_f = 220 \div 110$ ✓ $= 2$ A ✓ (2)

4.2.2 $I_a = I_L + I_f = 40 + 2$ ✓ $= 42$ A ✓ (2)

4.2.3 $E = V + I_a R_a$ ✓ $= 220 + 42 \times 0{,}2$ ✓ $= 228{,}4$ V ✓ (3)

4.3 Increase the speed of rotation of the armature ✓, so the conductors cut the flux faster ✓. Increase the field current (stronger magnetic field / more flux) ✓, so more flux is cut per second ✓. (Also accept more turns on the armature.) (4)

[15]

QUESTION 5

5.1 $1 \div C_T = 1 \div 4 + 1 \div 6 + 1 \div 12$ ✓ $= 6 \div 12$ ✓; $C_T = 2\ \mu F$ ✓ (3)

5.2 $C_T = 4 + 6 + 12$ ✓ $= 22\ \mu F$ ✓ (2)

5.3 $5 + 3 = 2 + I$ ✓; $I = 6\ A$ ✓ (2)

5.4.1 $R_P = (6 \times 3) \div (6 + 3) = 2\ \Omega$ ✓✓; $R_T = 4 + 2 = 6\ \Omega$ ✓ (3)

5.4.2 $I_T = 12 \div 6$ ✓ $= 2\ A$ ✓ (2)

5.4.3 $V_{R1} = 2 \times 4$ ✓ $= 8\ V$ ✓ (2)

5.4.4 $V_P = 12 - 8 = 4\ V$ ✓; $I_{R2} = 4 \div 6 = 0{,}67\ A$ ✓ (2)

5.5 In any closed loop, the sum of the voltage rises (emfs) ✓ equals the sum of the voltage drops ✓ (the algebraic sum of the voltages is zero). (2)

5.6 Any TWO ✓✓: smoothing the power supply; decoupling / removing noise near ICs and motors; timing circuits (e.g. 555 timer); filtering signals; storing energy for short bursts; debouncing switches. (2)

[20]

QUESTION 6

6.1 A primary cell cannot be recharged and is thrown away when flat ✓ – e.g. zinc-carbon or alkaline AA cell ✓. A secondary cell can be recharged many times ✓ – e.g. lithium-ion, lead-acid, NiMH ✓. (4)

6.2 $t = Ah \div I = 60 \div 4$ ✓ $= 15\ h$ ✓ (2)

6.3 $V_{ph} = V_L \div \sqrt{3}$ ✓ $= 400 \div \sqrt{3}$ ✓ $= 230{,}94\ V$ ✓ (3)

6.4 Any TWO ✓✓: more power with less copper (smaller conductors); three-phase motors are self-starting and run smoothly; two voltages available (400 V and 230 V); steadier power delivery; more efficient transmission. (2)

6.5 Axes labelled (V and I) ✓; forward region with current rising sharply after the knee ✓ at about 0,7 V labelled ✓; reverse region with a tiny leakage current and a sharp rise at the breakdown voltage labelled ✓ (4)

[15]

QUESTION 7

7.1.1 $V_{p\text{-}p} = 6 \times 2$ ✓ $= 12\ V$ ✓ (2)

7.1.2 $V_p = 12 \div 2 = 6\ V$ ✓ (1)

7.1.3 $T = 4 \times 5 \checkmark = 20 \text{ ms} \checkmark$ (2)

7.1.4 $f = 1 \div 0{,}02 \checkmark = 50 \text{ Hz} \checkmark$ (2)

7.2 X = (A + B) · C ✓✓; truth table ✓✓✓✓ (one mark for each two correct rows): (6)

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

7.3 Any TWO inputs ✓✓: push buttons, limit switches, proximity sensors, photoelectric sensors, thermocouples, encoders, pressure switches. Any TWO outputs ✓✓: contactors / motor starters, solenoid valves, lamps, buzzers, relays, VSD signals. (4)

7.4 Any THREE ✓✓✓: clean the pad and the component lead; heat the pad and the lead together with a clean, tinned iron for a second or two; feed the solder onto the joint (not onto the iron) until it flows around the lead; remove the solder and then the iron, and do not move the joint while it cools; a good joint is shiny and cone-shaped; cut off the extra lead. (3)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE